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AI-Enabled Cardiac Digital Twin from Wearable Sensors

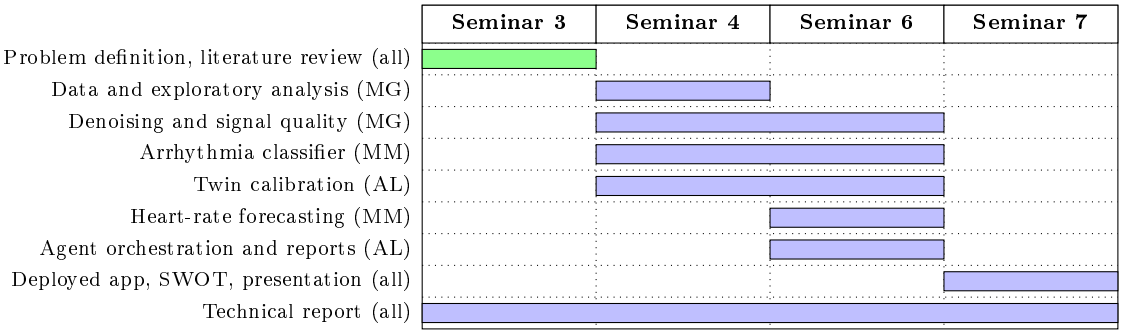
– Advanced Machine Learning report –

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Work plan (MG: Marc Grec, MM: Mihai Moldovan, AL: Adrien-Jeno Lorincz; green: done).

Abstract

Wearables record a single-lead ECG continuously, but the signal is noisy and differs from person to person. We propose a simple cardiac digital twin: a model of the patient's heartbeat, fitted to their own normal beats and shared by five AI agents that clean the signal, detect arrhythmias, forecast heart rate, update the twin and write a report for a clinician (Figure 2.1). Beats that differ strongly from the twin are flagged as unusual for that patient. We evaluate the agents on public PhysioNet ECG data, always on patients the models have not seen. Adding the twin's deviation score to a beat classifier raises the F1 score for ventricular beats from 0.88 to 0.96.

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1 Introduction

Cardiovascular diseases are the leading cause of death worldwide, with an estimated 19.8 million deaths in 2022 [20]. Smartwatches can already screen people for heart-rhythm problems, but a single-lead wearable ECG is noisy and limited on its own [1]. Digital twins, models of one patient that are updated with that patient's data, are a promising way to personalise monitoring [4]. However, current cardiac twins are mostly built from hospital data such as medical imaging [7].

We build a simple twin from wearable data alone: a heartbeat model [12] fitted to the wearer's own beats, combined with machine-learning models that clean the signal, detect arrhythmias and forecast heart rate. Five agents share the twin and write a report for a clinician. The twin is then used to measure how far each new beat is from the patient's own normal beat.

2 Scientific Problem

2.1 Problem definition

Informal description. Smartwatches and chest patches can record a single-lead ECG during everyday life, but these recordings are hard to use for three reasons:

- **Noise.** The person is moving, and wearable sensors have many sources of error, so the signal is often disturbed [1].
- **Volume.** A wearable records all day, every day, which is far more data than a clinician can read.
- **Every patient is different.** ECG shapes vary from person to person, so models trained on other people make more mistakes on a new patient [5, 10].

Our idea is a *cardiac digital twin*: a model of one patient's heart that is kept up to date with that patient's own data [4]. Existing cardiac twins are built from hospital imaging [7]; ours uses only wearable data. It is a simple mathematical model of a heartbeat [12], adjusted to the patient's own normal beats. When a new beat looks very different from what the twin expects, it is flagged as unusual for this patient. Five agents share the twin: Signal Monitoring, Arrhythmia Detection, Physiological Prediction, Twin Calibration and Clinical Recommendation. The result is a short report for a doctor; the system does not make diagnoses.

Why an intelligent algorithm? Fixed rules fail when the signal is noisy or the patient is atypical, while deep networks trained on large datasets classify heart rhythms from single-lead ECG as well as cardiologists [9]. Learning is also needed to clean the signal, forecast heart rate and fit the twin to each patient.

2.2 Formal problem statement

The input is a noisy ECG signal from a wearable, possibly together with heart rate and motion:

$$\tilde{x}(t) = x(t) + \eta(t), \tag{2.1}$$

where x is the true heart signal and η is the noise. From $\tilde{x}$ we cut out the individual heartbeats $b_1, b_2, \ldots$

The twin describes a heartbeat by its five waves (P, Q, R, S, T), each with a position, a height and a width, i.e. 15 numbers ϑ_p for patient p [12]. Calibrating the twin means choosing ϑ_p so that the beats it generates match the patient's normal beats. For each new beat b_k the twin predicts a beat $\hat{b}_k$, and the *deviation score*

$$d_k = \|b_k - \hat{b}_k\| \tag{2.2}$$

measures how unusual the beat is for this patient. A typical beat gives a small score, while a beat the twin cannot reproduce, for example an extra beat coming from the ventricles, gives a large one. Table 2.1 lists the input and output of each agent.

Table 2.1: Input and output of each agent.

Agent	**Input**	**Output**
Signal Monitoring	raw ECG	cleaned ECG and a signal-quality score
Arrhythmia Detection	heartbeats and their deviation scores	a label for every beat: normal (N), supraventricular (S), ventricular (V), fusion (F) or unknown (Q) [5]
Physiological Prediction	recent heart rate	heart-rate forecast
Twin Calibration	clean normal beats	updated twin ϑ_p and deviation scores d_k
Clinical Recommendation	outputs of all agents	report for the clinician

The agents work together as a pipeline (Figure 2.1). The Signal Monitoring agent cleans the signal and marks segments that are too noisy to use. The Arrhythmia Detection agent labels every beat, using the deviation score as an extra hint. The Physiological Prediction agent forecasts the heart rate. The Twin Calibration agent keeps the twin up to date, using only clean beats that were labelled normal.

Finally, the Clinical Recommendation agent collects these results into a short report and highlights anything a doctor should look at.

2.3 Graphical abstract

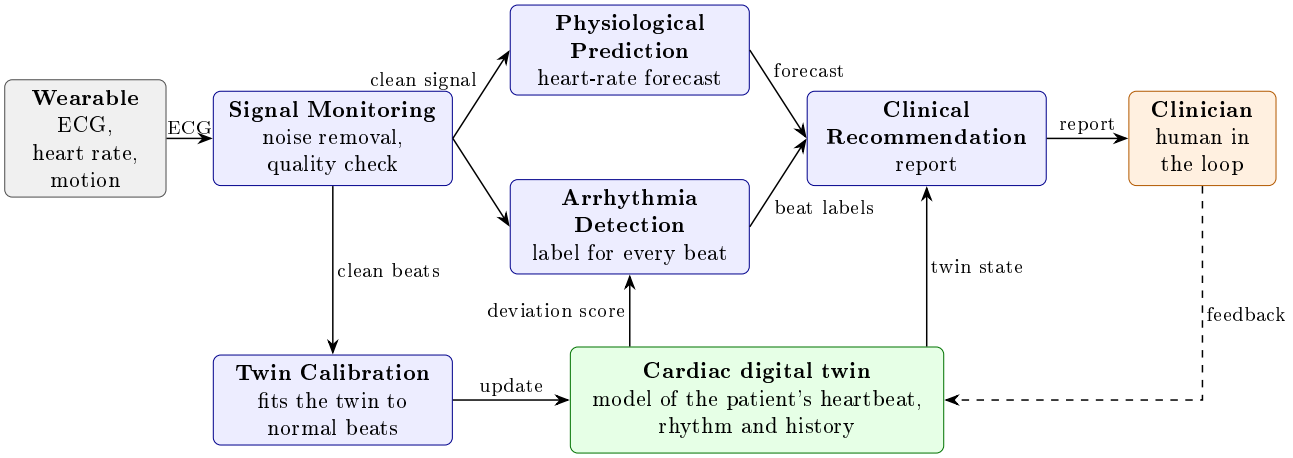


Figure 2.1: Graphical abstract: five agents share a model of the patient's heart (the twin); the clinician's feedback updates it.

3 State of the art/Related work

We searched Crossref, Semantic Scholar, OpenAlex, PubMed and the web for work in the areas below.

3.1 Cardiac digital twins

Corral-Acero et al. [4] define a healthcare digital twin as a model that combines a person's data over time, with wearables as one of its data sources. They argue that combining physiological models with statistical models learned from data is the way towards precision cardiology.

Most implemented cardiac twins use hospital data. Gillette et al. [7] build the heart's anatomy from MR images in a few hours per patient and fit its electrical behaviour to a 12-lead ECG, which they demonstrated for a single subject. Simonian et al. [18] build a twin of the mitral valve that integrates the patient's follow-up data to predict the valve's state after a repair procedure. Thangaraj et al. [19] describe hybrid twins that combine physiological models with lightweight machine learning. They see smartwatches as a way to close the loop with the patient, but note that this only works if people keep wearing them. Closest to our work, Roop et al. [16] model the heart's electrical conduction system and personalise it with wearable signals, so that it produces ECGs that match the individual.

Our twin needs only a wearable and is used for continuous monitoring.

3.2 Wearable ECG and signal quality

Bayoumy et al. [1] review wearables in cardiovascular care. They find that a single-lead ECG is enough only for the simpler arrhythmias, and list accuracy, privacy, cost and separating useful information from noise as the main barriers to adoption. Bhavsar et al. [2] treat wearable measurements as digital biomarkers and propose a platform that connects patients and doctors and adds insights from continuous monitoring to clinical reports. Del-Valle-Soto et al. [6] survey wearable health applications, describe how they enable real-time feedback and personalised interventions, and stress the importance of privacy and data security.

Noise is the main technical obstacle. Sameni et al. [17] place a mathematical model of the ECG, the same kind of model our twin is based on, inside a Kalman filter that follows the signal over time. This removed noise better than standard filters over a wide range of noise levels.

We compare a standard filter with a learned denoiser on real recorded noise, with the aim of measuring how much cleaning helps arrhythmia detection.

3.3 Arrhythmia classification

De Chazal et al. [5] split the MIT-BIH Arrhythmia Database into two groups of patients, one for training and one for testing, which became the reference way to evaluate beat classifiers. With this split, their classifier detected some beat types, especially supraventricular beats, poorly. Kiranyaz et al. [10] therefore train a small network for each patient, using common data plus the patient's own labelled beats. It is fast enough for real-time monitoring and, according to the authors, could run on a lightweight wearable device. Hannun et al. [9] trained a deep network on single-lead ambulatory ECG that recognises 12 rhythm classes as well as cardiologists. They see its role in triage, with results confirmed by a clinician.

We keep one general model and add the twin's deviation score as a feature; the score needs no labelled data from the new patient.

3.4 Heart models and forecasting

McSharry et al. [12] proposed the heartbeat model behind our twin, originally to generate realistic synthetic ECG for testing signal-processing methods. The model is controlled by a few intuitive

settings: the heart rate, how much it varies, and the shape of the P, Q, R, S and T waves.

For forecasting heart rate, the Temporal Fusion Transformer [11] predicts several steps ahead and shows which inputs matter most. It combines different kinds of input, such as fixed information and past measurements, and was more accurate than earlier methods on real-world datasets.

We fit the heartbeat model to each patient and use it as a personal reference beat.

3.5 AI agents in healthcare

PHIA [13] is a language-model agent that analyses wearable health data by writing code and looking up information. In an evaluation by experts, it answered 84% of factual numerical questions correctly.

In our system the agents coordinate the models and write a report, and the decisions stay with the clinician.

4 Investigated approach

This chapter describes the pipeline and its first results. All experiments use public PhysioNet data [8]. Models are always trained on one group of patients and tested on a different group, so every result describes patients the models have never seen.

4.1 Pipeline

Figure 2.1 shows how the agents connect. Each agent currently has a simple version that is evaluated on its own (Table 4.1). Connecting the agents and generating the clinical report are left for future work.

Table 4.1: Current version of each agent.

Agent	Method	Code
Signal Monitoring	band-pass filter vs. convolutional denoising autoencoder	`denoise.py`
Arrhythmia Detection	gradient-boosted trees on RR and shape features vs. a 1-D CNN	`classify.py`
Physiological Prediction	last value (persistence) vs. a linear autoregressive model	`forecast.py`
Twin Calibration	McSharry model fitted to the patient's first 5 minutes	`twin.py`
Clinical Recommendation	future work	–

4.2 Data and exploratory analysis

Data. We use two PhysioNet databases. The MIT-BIH Arrhythmia Database contains 48 half-hour, two-channel ECG recordings from 47 subjects, sampled at 360 Hz, with every beat labelled by cardiologists [14]; we use the MLII lead. The MIT-BIH Noise Stress Test Database contains recorded ambulatory noise of three types: baseline wander, muscle artefact and electrode motion [15]. Following de Chazal et al. [5], we split the recordings by patient: 22 recordings for training (DS1) and 22 for testing (DS2), leaving out the four recordings of patients with pacemakers. Records 201 and 202 come from the same subject [14], so we drop 202 from the test set to keep the two groups of patients completely separate. Each beat is cut out from 250 ms before to 450 ms after its R peak, after a 0.5–40 Hz band-pass filter, and labelled with one of the five standard classes N, S, V, F and Q (Table 4.2).

Table 4.2: Beats per class in the training and test patients.

	N	S	V	F	Q	Total
Training (DS1, 22 patients)	45 824	943	3 788	414	8	50 977
Test (DS2, 21 patients)	42 159	1 781	3 200	387	7	47 534

What the data show. About 90% of all beats are normal, so accuracy alone would be misleading and we report results per class. Figure 4.1 shows the average beat of each class: ventricular (V) beats have a wide, deformed QRS complex, while supraventricular (S) beats look almost like normal ones. What distinguishes S beats is their timing: their median preceding RR interval is 82% of the local average, compared with 73% for V beats and 100% for normal beats (Figure 4.2).

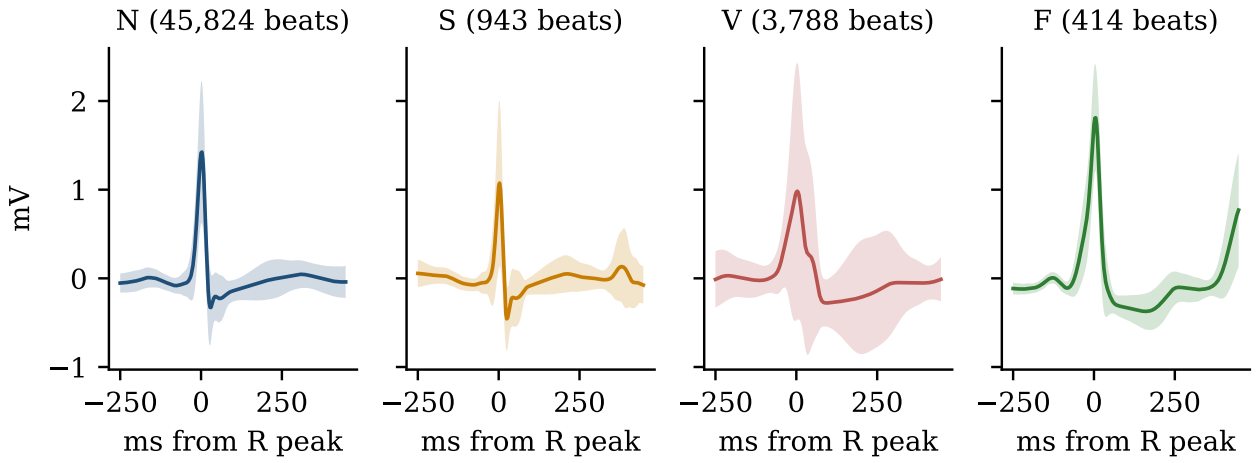


Figure 4.1: Average beat (± one standard deviation) of each class in the training patients.

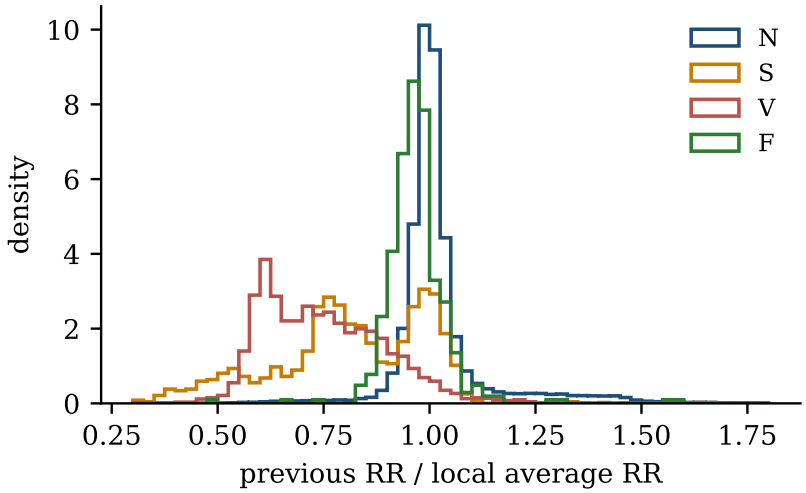


Figure 4.2: Timing of each beat relative to the local rhythm: premature beats (S, V) arrive early.

Figure 4.3 shows why a personal model is useful. Across the 43 patients, a normal beat differs from the same patient's typical beat by 0.058 mV (median RMS), but typical beats of two different patients differ by 0.252 mV, more than four times as much. A model of a normal beat is therefore much more precise when it is built for one patient. Figure 4.4 shows the three noise types we add to test the Signal Monitoring agent.

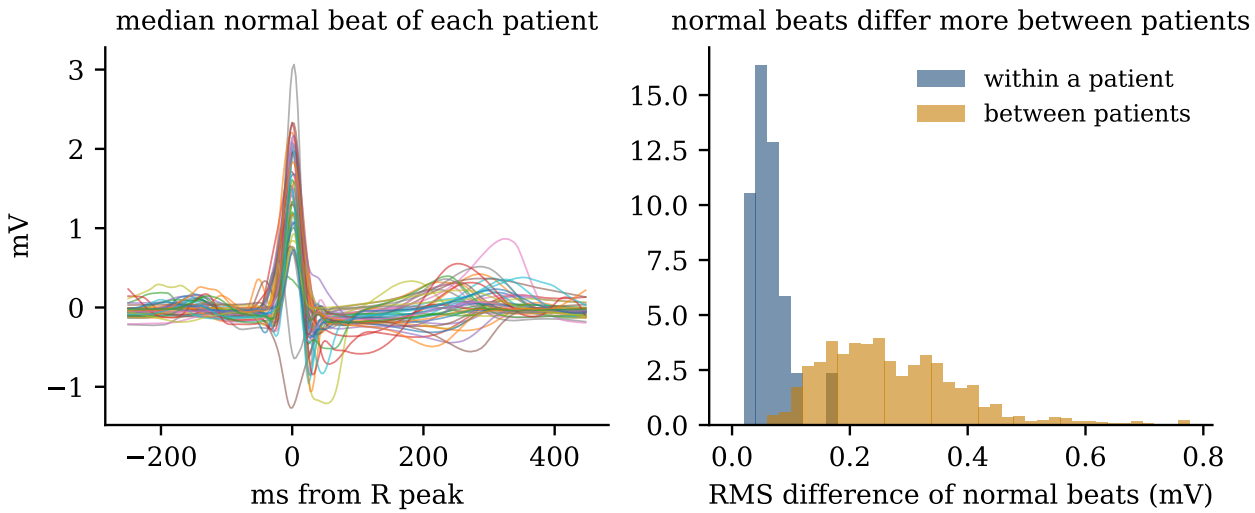


Figure 4.3: Left: the typical normal beat of each of the 43 patients. Right: normal beats vary much less within one patient than between patients.

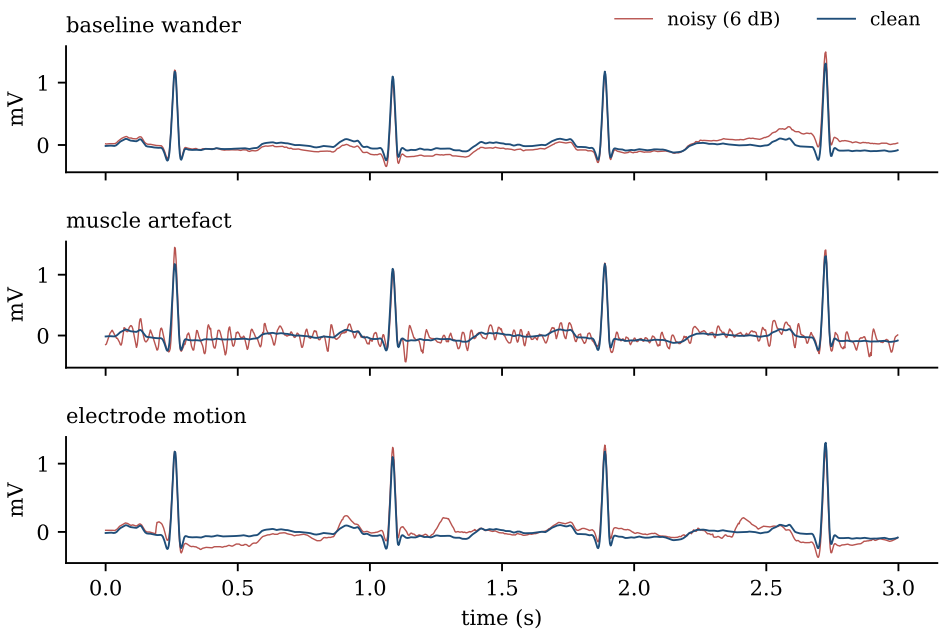


Figure 4.4: A clean ECG segment with each recorded noise type added at a signal-to-noise ratio of 6 dB.

4.3 Signal Monitoring: noise removal

Method. We compare a standard 0.5–40 Hz band-pass filter with a convolutional denoising autoencoder, a neural network that learns to map a noisy segment to the clean one [3]. Our autoencoder compresses 1.4-second segments with three convolution layers and reconstructs them with three transposed convolutions. It is trained on segments from the training patients mixed with noise from the first half of each noise recording, at random signal-to-noise ratios between −6 and 18 dB. Testing uses the test patients and the second half of the noise recordings, so neither the patients nor the noise are seen in training. We measure the improvement in signal-to-noise ratio (SNR).

Table 4.3: SNR improvement (dB) on the test patients; higher is better, best per row in bold.

Noise type	Input SNR	Band-pass filter	Autoencoder
Baseline wander	0 dB	**12.4**	9.8
	6 dB	**10.4**	5.3
	12 dB	**7.6**	0.1
Electrode motion	0 dB	1.6	**7.2**
	6 dB	1.3	**3.8**
	12 dB	**0.6**	−0.7
Muscle artefact	0 dB	4.8	**9.0**
	6 dB	4.3	**4.9**
	12 dB	**2.9**	0.1

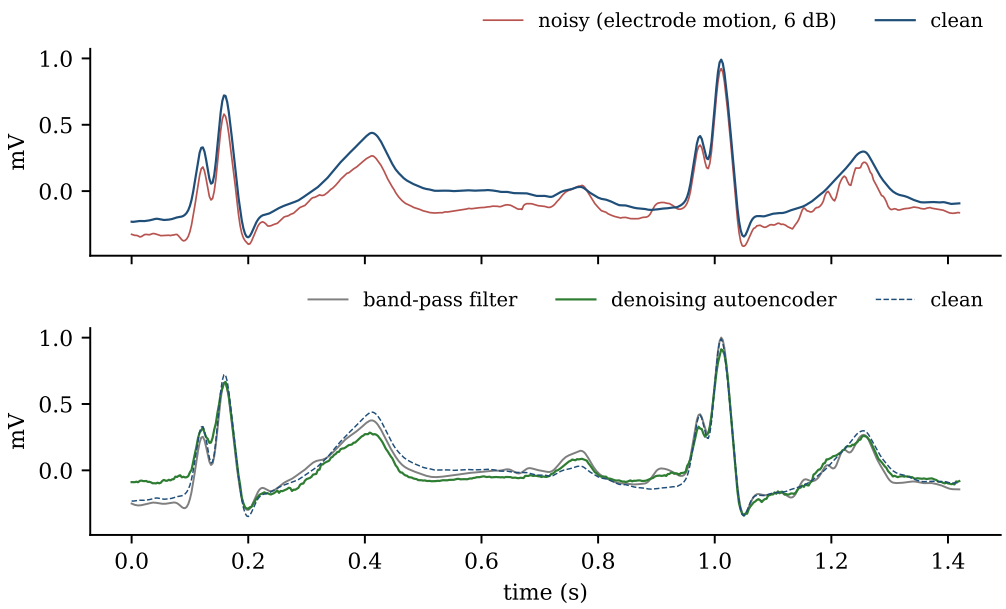


Figure 4.5: Electrode-motion noise at 6 dB on a test patient (top), and the output of both methods (bottom).

Results. The two methods are complementary (Table 4.3). The band-pass filter is best for baseline wander, which lies mostly below its pass band. The autoencoder is much better for strong electrode-motion and muscle noise, which overlap with the ECG in frequency and which a fixed filter cannot remove: at 0 dB it gains 7.2 dB on electrode motion, against 1.6 dB for the filter (Figure 4.5). On mildly noisy signals (12 dB), however, the autoencoder adds small distortions of its own. A combined approach would therefore filter first and apply the autoencoder only when the signal-quality check reports strong noise.

4.4 Arrhythmia Detection: beat classification

Method. Each beat is described by its timing (previous and next RR interval, the local average of the last 10 RR intervals, and the two ratios to it) and its shape. We compare two classifiers. The baseline is a gradient-boosted tree model on the timing features plus a downsampled beat shape, similar to de Chazal et al. [5]. The second is a small 1-D convolutional neural network that reads the whole beat and combines it with the timing features [10], trained for 8 epochs with three different random seeds. Both use class weights so that rare classes count as much as normal beats. Models are trained on all training patients and tested on the test patients' beats after their first 5 minutes, the same beats used in the twin experiments (39 623 beats).

Table 4.4: Beat classification on unseen patients: sensitivity (Se), positive predictivity (+P) and F1 per class. CNN values are the mean of three seeds.

	Baseline (boosted trees)			**1-D CNN**		
Class	Se	+P	F1	Se	+P	F1
N (normal)	0.91	0.96	**0.93**	0.86	0.98	0.92
S (supraventricular)	0.17	0.32	0.22	0.62	0.27	**0.37**
V (ventricular)	0.94	0.82	**0.87**	0.89	0.73	0.80
F (fusion)	0.31	0.03	0.06	0.32	0.04	**0.08**
Macro-F1			0.52			**0.54** ($\pm$ 0.003)

Results. Normal and ventricular beats are recognised well even on unseen patients (Table 4.4). Supraventricular beats remain the hardest class, as de Chazal et al. also found [5]: the baseline finds only 17% of them. The CNN finds 62%, but at the cost of more false alarms, and it is weaker than the baseline on ventricular beats. Fusion beats are rare and resemble normal beats, so neither model detects them reliably yet. Section 4.6 tests whether the twin's deviation score helps.

4.5 Physiological Prediction: heart-rate forecasting

Method. From the last 20 RR intervals we forecast two targets: the next RR interval, and the average RR interval of the next 10 beats, i.e. the heart rate over the next few seconds. We compare persistence ("the next value equals the last one", or the average of the last 10 for the 10-beat target) with a linear autoregressive model. The model sees the past intervals relative to their own average, so a single model fits slow and fast hearts alike. It is trained on the training patients and tested on the test patients.

Table 4.5: Mean absolute error of RR forecasts on the test patients (milliseconds; lower is better).

Target	**Persistence**	**Autoregressive**
Next RR interval	113.0	**87.4**
Average of the next 10 RR intervals	37.5	**35.5**

Results. The autoregressive model improves the next-beat forecast (Table 4.5), probably because it can learn that a premature beat is usually followed by a longer pause, which persistence cannot. For the 10-beat average both methods are close, since the heart rate changes slowly over a few seconds. These two models serve as baselines for more complex forecasters such as the temporal fusion transformer [11].

4.6 Twin Calibration and the deviation score

Method. For every patient, the twin is calibrated on the first 5 minutes of the recording, without using any labels: we take the patient's median beat and fit the 15 wave parameters of the McSharry model [12], together with a gain and an offset, by nonlinear least squares. On synthetic beats with known parameters, this procedure recovered the true beat to within the added noise in all nine tests, at three different heart rates. For each later beat, the deviation score d_k compares the beat with the twin's beat, relative to the patient's own beat-to-beat variation, so a typical beat has $d_k \approx 1$. As a control we compute the same score against the patient's raw median beat, i.e. without the physiological model.

Calibration. Across the 43 patients, the calibrated twin reproduces the patient's median beat with a median error of 0.032 mV, against 0.181 mV for the uncalibrated population twin (Figure 4.6, left).

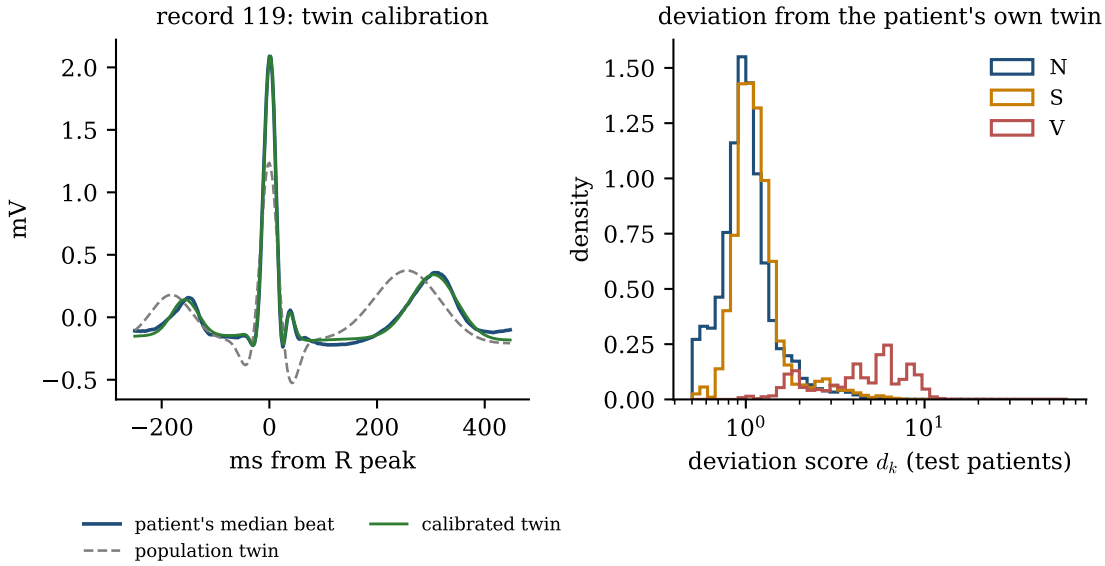


Figure 4.6: Left: twin calibration for one patient. Right: deviation scores of all test beats after calibration; ventricular beats stand out, supraventricular beats do not.

Deviation score. On the test patients, normal beats have a median deviation of 1.07 and ventricular beats 6.09 (Figure 4.6, right). The score alone separates ventricular from normal beats with an AUROC of 0.98, but supraventricular beats only weakly (0.60), because their shape is almost normal. The model-free control scores the same for ventricular beats (0.98) and slightly lower for supraventricular ones (0.57).

Does the twin help the classifier? We add the deviation score as an extra feature to the baseline classifier and compare it with the same classifier without it. For a fair comparison, both are trained only on beats after the first 5 minutes, which is why the baseline differs slightly from Table 4.4. Since each patient is one independent unit, we give 95% intervals from a bootstrap over the 21 test patients (Table 4.6).

Table 4.6: Effect of the deviation score on the baseline classifier (test patients). Differences are relative to the classifier without the score, with 95% bootstrap intervals over patients.

	Without score	**+ twin score**	**+ control score**
Macro-F1	0.538	0.561	0.537
difference	–	+0.036 [−0.021, 0.110]	+0.008 [−0.031, 0.056]
F1, ventricular (V)	0.879	0.956	0.943
difference	–	+0.085 [0.022, 0.205]	+0.073 [0.017, 0.191]
F1, supraventricular (S)	0.309	0.224	0.216
difference	–	−0.060 [−0.177, 0.097]	−0.070 [−0.199, 0.101]

Discussion. The twin's score helps for beats whose shape differs from normal: ventricular F1 rises from 0.88 to 0.96, and the 95% interval of the gain excludes zero. The overall macro-F1 also rises, but with only 21 test patients the interval still includes zero, so we cannot yet claim an overall improvement. Supraventricular beats are not helped, as expected from Section 4.2: they differ from normal beats mainly in timing. On detection alone, the physiological twin performs about as well as the model-free control. Its added value is that its parameters have a meaning (for example the width of the QRS complex) and that it can simulate the patient, which is useful for the clinical report. Next steps are to combine the score with the CNN, add timing to the twin, and test on a wearable dataset.

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